

Aryan Singh

Machine Learning Engineer | Data Scientist

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PROFILE SUMMARY

Machine Learning Engineer and **Data Scientist** with practical experience in **Machine Learning**, **Deep Learning**, **Data Analysis**, and **Flask Application Development**. Proficient in **Python**, **SQL**, and **Predictive Modeling**, with expertise in developing data-driven solutions and scalable applications. Skilled in building end-to-end AI solutions, from data analysis and model training to deployment using **Flask/FastAPI**. Passionate about leveraging **Artificial Intelligence**, **Data Science**, and **Software Engineering** to deliver impactful solutions.

SKILLS

Python, SQL (MySQL), Tableau

Data Cleaning, Data Analysis, Data Visualization, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation, Hyperparameter Tuning, Classification, Regression, Neural Networks, Machine Learning, Deep Learning, PyTorch, Scikit-learn, NLP (spaCy), Flask, FastAPI, Git, GitHub

INTERNSHIP EXPERIENCE

Certify Technology

Nov 2025 - Dec 2025

Web Developer intern

Remote, India

- Developed and maintained **backend web applications** using **Python**, **Flask**, and **SQL databases**.
- Designed and integrated **RESTful APIs** to enable secure and efficient communication between application components.
- Implemented database operations, data validation, and business logic to support scalable web solutions.
- Optimized application performance and collaborated with cross-functional teams to deliver reliable and user-centric software.

Rogue Code

July 2025 - July 2025

DS & DL intern

Mumbai, India

- Designed and implemented end-to-end **Machine Learning** and **Deep Learning** workflows, covering **data preprocessing**, **feature engineering**, and **model evaluation**.
- Built and trained **PyTorch-based neural networks**, including **CNN** and **LSTM** architectures, for classification and prediction tasks.
- Conducted model validation, performance analysis, and **hyperparameter tuning** to enhance accuracy and generalization.
- Utilized **Python**, **PyTorch**, and **Scikit-learn** to develop scalable AI solutions for real-world applications.

Projects

Swar-Samved: AI Music Tutor

May 2026

Python | FastAPI | Librosa | React | Next.js | TypeScript

Personal Project

- Developed an **AI-powered vocal analysis platform** that evaluates **pitch**, **rhythm**, and **tonal accuracy** in real time using advanced audio signal processing techniques.
- Implemented **audio feature extraction**, **spectrogram analysis**, and **PYIN-based pitch detection** using **Librosa**, **NumPy**, and **SciPy** to generate intelligent performance insights.
- Built a full-stack application with **FastAPI**, **React**, **Next.js**, and **Web Audio API** for real-time audio recording, visualization, and personalized feedback generation.

Human Activity Recognition System

July 2025

Python | PyTorch | LSTM | Deep Learning

Internship Project

- Developed a **Human Activity Recognition (HAR)** system using **LSTM-based Deep Learning** to classify human activities from time-series sensor data with high accuracy.
- Performed **data preprocessing**, **feature engineering**, and sequence generation, followed by model training and evaluation using PyTorch.
- Evaluated model performance using **accuracy**, **precision**, **recall**, and **confusion matrix metrics**, enabling efficient real-time activity prediction.

EDUCATION

Vidyavardhini's College of Engineering and Technology

2023-2027

B.E. Computer Science Engineering (Data Science)

Vasai, Palghar

CERTIFICATIONS

- Coursera - IBM Data Science Professional Certificate
- Coursera - IBM Machine Learning Professional Certificate